APPROACHES TO CONTAINERS OVER A BASE: EXTENSIVITY, CARTESIAN CLOSURE, AND THE LINEAR COLLAPSE

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Abstract. The phrase "a container in a category" names at least four inequivalent constructions: external families over a base $\mathcal{C}$, indexed dependent-polynomial functors over a locally cartesian closed base, fibrational containers over a comprehension category, and Walker's locally subcartesian closed weakening. Over **Set** they coincide; off **Set** they diverge, and no map has been drawn of *where*. We draw one — a census of the bases over which the construction has been run, from toposes down through Weber's and Walker's weakenings to additive bases — and argue that two properties of the base — extensivity and local cartesian closure — are the two axes of divergence, and we use the category of vector spaces, which has neither, as the discriminating example. Only the external-family construction $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ reaches $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$; the other three cannot even form their semantics there. We make this quantitative with three theorems about $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$: the container extension is fully faithful iff the monoidal unit is connected (*not* iff $\mathcal{C}$ is extensive); the Dirichlet tensor is closed iff a familial-representability condition holds, which over a linear base isolates the finite-dimensional corner; and the substitution product is left-closed exactly where it collapses onto the Dirichlet tensor — so that extensivity, the obstruction to the first two, is what would have to fail for the third. The upshot is a two-axis classification in which $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ forces the external approach. Along the way we develop the fibrational approach beyond its refereeing role into an explicit *logic of containers*: applying the container construction to the codomain fibration of **Set** yields a bifibration whose fibred quantifiers are the All and Exists predicate liftings, and whose hyperdoctrine is the fibrewise opposite of the usual one — container-$\exists$ is **Set**-$\forall$, and each fibre is a co-topos.

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Draft — work in progress; Kodamai grant deliverable. This revision adds the implementation-side "parallel program" motivation to §3. A further landscape item — the additive-lens cartesian-closed structure over commutative monoids as prior art, with this survey's linear-base obstructions as its negative complement — is deferred pending verification of its sources.

1. Introduction

1.1. **The ambiguity.** A *container* in the sense of Abbott, Altenkirch and Ghani [1] is a set S of shapes together with, for each shape s, a set P_s of positions; its extension is the polynomial endofunctor $[\![S,P]\!]X = \coprod_{s\in S} X^{P_s}$ on **Set**. Containers are the syntax of strictly positive datatypes, and their theory is by now classical: containers with their morphisms are equivalent to a full subcategory of $[\mathbf{Set},\mathbf{Set}]$, the substitution product models functor composition, directed containers are exactly (small) categories [2], and the whole package reappears inside Spivak's category **Poly** of polynomial functors [12].

The moment one asks for "a container in a category $\mathcal{C}$ other than **Set**," the theory forks. At least four constructions claim the name, and they are genuinely different functors:

(1) **External families.** An object of $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ is a *set* S of shapes together with position *objects* $P_s \in \mathcal{C}$; the extension $[\![S,P]\!] = \coprod_{s\in S}[P_s,-]$ uses the self-enrichment of a closed $\mathcal{C}$

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and an *external* set-indexed coproduct. This is the $\Sigma\Pi\mathcal{V}$ construction of Dorta, Jarvis and Niu [5].

(2) **Indexed / dependent-polynomial functors.** A diagram $I \leftarrow P \to S \to O$ *internal* to a locally cartesian closed base $\mathcal{C}$, with semantics the composite $\Sigma\,\Pi\,\Delta$ of slice functors — the polynomial functors of Gambino and Kock [8].

(3) **Fibrational containers.** Positions presented as a fibration over the shapes, with comprehension; this subsumes (1) and (2) as the family and codomain fibrations respectively, and, applied to the codomain fibration of **Set** itself, produces the bifibration *logic of containers.*

(4) **Subcartesian containers.** Walker's locally subcartesian closed bases [15], where an *affine* base-change $\nabla_f \dashv \boxtimes_f$ replaces the top adjoint of the $\Sigma \dashv \Delta \dashv \Pi$ chain when the base fails to be locally cartesian closed.

Over $\mathcal{C} = \textbf{Set}$ these are one construction viewed four ways. Over a general base they are four constructions, and the literature offers no systematic answer to the question that then becomes unavoidable: *for which bases do they agree, and along what axis do they diverge?*

1.2. **The thesis.** We propose a two-property answer. Call a base *good* if it is both *extensive* (coproducts are disjoint and universal — equivalently $\mathcal{C}/(A+B) \simeq \mathcal{C}/A \times \mathcal{C}/B$) and *locally cartesian closed* (every slice is cartesian closed, so pullback has a right adjoint Π). The claim is:

> *Extensivity and local cartesian closure of the base are the two axes along which the four notions of container diverge. When the base has both, all four agree; and each approach fails to reach a base precisely when that base fails one of the two properties that the approach silently assumes.*

The sharpest instrument for separating the approaches is a base that fails *both* properties. The category **Vec** of vector spaces over a field k is exactly such a base: it is not extensive (the canonical comparison $A \amalg B \to A \oplus B$ from the coproduct-of-underlying-sets to the biproduct is not an isomorphism — "$\amalg \subsetneq \oplus$"), and it is not locally cartesian closed (there is no internal dependent product Π). We therefore run $\mathrm{Fam}(\textbf{Vec}^{\mathrm{op}})$ — linear containers — as the worked example throughout, and it does the discriminating work summarised in Table 1.

Base $\mathcal{C}$	extensive?	LCC?	(1) $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$	(2) $\Sigma\Pi\Delta$	(4) subcartesian
Set, any topos	✓	✓	reaches it	forms	(LSCC)
Vec (fd or gen.)	$\times$ ($\amalg \subsetneq \oplus$)	$\times$ (no Π)	**reaches it**	**cannot form** Π	**not LSCC**

TABLE 1. The discriminator. Over a base that is extensive and locally cartesian closed, all approaches coincide (the classical convergence: containers $\simeq$ directed containers $\simeq$ **Poly**-comonoids $\simeq$ categories). Over **Vec**, which has neither property, only the external approach (1) survives. Approach (3) sits above the table: it presents (1) and (2) as two fibrations and locates their divergence exactly where the base fails the two properties. This is the honest reason $\mathrm{Fam}(\textbf{Vec}^{\mathrm{op}})$ "needs the first approach."

1.3. **The quantitative face.** The discriminator is not merely a table of yes/no answers. The obstructions have exact measures, and supplying them is the technical contribution of the survey. Three theorems about the external approach (1), each proved in Section 4 in our own words, turn the two-axis classification into precise dichotomies. Write $[\![-]\!]\colon \mathrm{Fam}(\mathcal{C}^{\mathrm{op}}) \to \mathcal{C}\text{-}[\mathcal{C}, \mathcal{C}]$ for the container extension into $\mathcal{C}$-enriched endofunctors.

Theorem 1.1 (Fullness; T1, Theorem 4.4)**.** *$[\![-]\!]$ is fully faithful if and only if the monoidal unit I is* connected*, i.e. the functor $\mathcal{C}(I,-)\colon \mathcal{C} \to \mathbf{Set}$ preserves small coproducts. This is* not *extensivity of $\mathcal{C}$: the base $\mathbf{Set} \times \mathbf{Set}$ is extensive yet its unit is disconnected and $[\![-]\!]$ is not full, while over $\mathbf{Vec}$ the underlying-set functor $\mathcal{C}(k,-)$ fails to preserve $\oplus$ — so the unit is disconnected in this sense — and $[\![-]\!]$ is neither full nor faithful.*

Theorem 1.2 ($\otimes$-closedness; T2, Theorem 4.10)**.** *The Dirichlet tensor $\otimes$ on $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ is closed if and only if a familial-representability functor Φ is representable. Over a cartesian base this always holds; over the linear base it holds* only *on the corner $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ of finite shape-sets and finite-dimensional positions, failing over both $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ and $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ by dual mechanisms.*

Theorem 1.3 ($\lhd$-left-closedness; T4, Theorem 4.17)**.** *The substitution product $\lhd$ on $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ is left-closed exactly where it* collapses onto $\otimes$*: over a base whose positions are dualizable (*tiny*), $\lhd = \otimes$, and the left internal hom is the Dirichlet hom of Theorem 1.2. Consequently $\lhd$ is left-closed on $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ and obstructed over every extensive base. Extensivity is* opposed *to $\lhd$-left-closedness: the property whose failure costs fullness (T1) and $\otimes$-closedness (T2) is exactly what buys $\lhd$-closedness.*

Theorems 1.1–1.3 are the quantitative shadow of the discriminator: the fullness and closedness obstructions over **Vec** are precisely the base failing the extensive/LCC hypotheses that approaches (2)–(4) would need at the outset. The inversion in Theorem 1.3 — extensivity as villain for T1/T2 and hero-of-the-obstruction for T4 — is the survey's most surprising structural fact and the reason the classification is genuinely two-dimensional rather than a single "goodness" scale.

1.4. **What is and is not new.** The tensor $\otimes$ over a general base, a composition product $\lhd_{\mathrm{DJN}}$, and the theorem that $\lhd_{\mathrm{DJN}}$-comonoids are enriched categories (their Theorem 4.3; their Definition 4.2 is the definition of enriched cofunctor), are due to Dorta, Jarvis and Niu [5]; we cite these as the definitions our theorems are *about*, and re-prove nothing of theirs. [1] The **Set** fullness theorem is Abbott–Altenkirch–Ghani [1]; our contribution is to name its hidden hypothesis. The reconstruction of a container from an abstract familially-representable functor is Diers' [4], and uses extensivity of the codomain, not of the base — a distinction Section 4 turns into the resolution of the "extensivity" folklore. Day convolution is [3]; polynomial functors over LCC bases are [8]; the composition-monoid refinement of the indexed corner is De Pascalis–Uustalu–Veltrì [6]; the subcartesian weakening is Walker [15]. In Section 6 the facts that Fam preserves fibrations and that cod is a bifibration are folklore [10, 11], and the fibrewise-op presentation $\mathbf{Cont} = \int_{\mathbf{Set}} \mathrm{cod}^{\mathrm{op}}$ is von Glehn's [13]. The new material is the two-axis discriminator, the census of Section 3 together with the assembly of the individually-known weakenings (Gambino–Kock, Weber, Walker) into one totally ordered tower and the placement of a given base on it, the three theorems T1/T2/T4 with their named invariants, their assembly into a single classification with $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ as the forcing example, and — for approach (3) — the explicit assembly of $\mathbf{Cont}(\mathrm{cod})$ as the bifibration of proof-relevant predicates on positions, the identification of its fibred quantifiers with the $\mathsf{All}/\mathsf{Exists}$ liftings, and the dualisation reading that makes the container hyperdoctrine the fibrewise opposite of the standard one.

[1] **Scope correction (2026-08-31).** Under the identification $\Sigma\Pi\mathcal{C} = \mathrm{Fam}(\mathcal{D}^{\mathrm{op}})$, $\mathcal{D} = (\Pi\mathcal{C})^{\mathrm{op}}$, our $\otimes$ agrees with theirs exactly, but their composition product does *not* agree with ours: at a position $(i, j\colon A_i \to J)$ their direction object is $\prod_{a\in A_i}\prod_{b\in B_{ja}}(u_{i,a}\cdot v_{ja,b})$ (their Def. 3.5 / Lemma 3.6) — the *outer* u multiplied in — whereas ours is $\prod_{a\in A_i}\prod_{b\in B_{ja}} v_{ja,b}$. The two coincide exactly at $\mathcal{C} = 1$. A smallest witness: $\mathcal{C} = 2$ with $\cdot = \wedge$, $e = \top$, $p = \sum_{i\in 1}\prod_{a\in 1}\bot$ and $q = \sum_{j\in 1}\prod_{b\in 1}\top$ give $\bot$ for theirs and $\top$ for ours. Moreover our defining property $[\![p \lhd q]\!] \cong [\![p]\!] \circ [\![q]\!]$ is unavailable on $\Sigma\Pi\mathcal{C}$ for $\mathcal{C} \neq 1$, since their embedding lands in presheaves on $\Pi\mathcal{C}$ rather than in endofunctors. We therefore write $\lhd_{\mathrm{DJN}}$ for theirs and $\lhd$ for ours, and attribute to [5] the general-base construction, not a composition-representing $\lhd$ over a general base.

1.5. **Outline.** Section 2 fixes the base and the two tensors. Section 3 maps the territory: a census of the bases over which the construction has been run, the weakening tower that orders those between **Set** and **Vec**, and the organising fact that off the good zone a base fails for exactly one of two reasons. Section 4 develops the external approach (1) and proves T1, T2, T4. Section 5 treats the indexed/dependent- polynomial approach (2) and shows why it cannot form its semantics over **Vec**. Section 6 treats the fibrational approach (3): first as the referee that presents (1) and (2) as two fibrations, then as a construction in its own right, the bifibration **Cont**(cod) whose dualised quantifiers are the logic of containers. Section 7 treats Walker's approach (4) and records the verdict that **Vec** is not locally subcartesian closed. Section 8 assembles the discriminator; Section 9 names the single sharpest open problem the survey surfaces.

2. Preliminaries

We collect only what the later sections use. Throughout, $(\mathcal{C}, \otimes, I, [-,-])$ is a *closed symmetric monoidal category with small coproducts* $\coprod$. Closedness gives the enrichment of $\mathcal{C}$ over itself, so that $\mathcal{C}(A,B) = \mathcal{C}(I,[A,B])$ naturally, and the adjunction $-\otimes Q \dashv [Q,-]$.

Definition 2.1 (Container over $\mathcal{C}$)**.** $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ is the category with objects $(S,(P_s)_{s\in S})$, where S is a set of *shapes* and each $P_s \in \mathcal{C}$ is a *position object*, and morphisms $(S,P) \to (T,Q)$ the pairs $(f,(\varphi_s))$ with $f\colon S \to T$ a function and $\varphi_s\colon Q_{f(s)} \to P_s$ in $\mathcal{C}$ (contravariant on positions). Explicitly,

$$\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})\big((S,P),(T,Q)\big) \;=\; \textstyle\prod_{s\in S}\ \coprod^{\mathbf{Set}}_{t\in T}\ \mathcal{C}(Q_t,P_s),$$

a choice of one $t = f(s)$ per shape plus a position map. We write $\langle Z\rangle := (\{*\}, Z)$ for the one-shape generator at $Z \in \mathcal{C}$; every object is the coproduct $(S,P) = \coprod_{s\in S}\langle P_s\rangle$, so $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ is the free coproduct completion of $\mathcal{C}^{\mathrm{op}}$.

Definition 2.2 (Extension)**.** The *extension* of (S,P) is the $\mathcal{C}$-enriched endofunctor

$$[\![S,P]\!] \;:=\; \textstyle\coprod_{s\in S}\,[P_s,-] \;:\; \mathcal{C} \longrightarrow \mathcal{C},$$

a coproduct of corepresentables. Over **Set** (with $I = 1$, $[P,X] = X^P$) this is the classical $[\![S,P]\!]X = \coprod_s X^{P_s}$; over **Vec** (with $I = k$, $[P,W] = \mathbf{Vec}(P,W)$, $\coprod = \oplus$) it is the linear extension $[\![S,P]\!]W = \bigoplus_s \mathbf{Vec}(P_s,W)$.

Definition 2.3 (The two tensors)**.** On $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ there are two monoidal structures whose extensions we use.

(1) The *Dirichlet* (parallel) tensor $(S,P)\otimes(T,Q) := (S\times T,(P_s\otimes Q_t))$, with unit $(\{*\}, I)$. Its extension is the Day convolution [3] of the corepresentables: $[\![(S,P)\otimes(T,Q)]\!] = \coprod_{s,t}[P_s \otimes Q_t,-]$, using $[P,-]*[Q,-] = [P\otimes Q,-]$.
(2) The *substitution* (composition) product $\triangleleft$, the non-symmetric monoidal structure whose extension is functor composition: $[\![(S,P)\triangleleft(T,Q)]\!] = [\![S,P]\!]\circ[\![T,Q]\!]$ [5, 12].

We use four properties of a base, each recalled with the only fact about it we need.

Definition 2.4. Let $\mathcal{C}$ be as above.

- $\mathcal{C}$ is *extensive* if finite coproducts are disjoint and universal; equivalently $\mathcal{C}/(A+B) \simeq \mathcal{C}/A \times \mathcal{C}/B$.
- $\mathcal{C}$ is *locally cartesian closed* (LCC) if every slice $\mathcal{C}/X$ is cartesian closed; equivalently every pullback functor f^* has a right adjoint Π_f (dependent product).
- The monoidal unit I is *connected* if $\mathcal{C}(I,-)\colon \mathcal{C} \to \mathbf{Set}$ preserves small coproducts.
- An object $Z \in \mathcal{C}$ is *tiny* if $[Z,-]\colon \mathcal{C} \to \mathcal{C}$ preserves small coproducts. In a closed symmetric monoidal category every *dualizable* object is tiny, since then $[Z,-] = Z^*\otimes(-)$ is a left adjoint. Over **Vec**, tiny = dualizable = finite-dimensional.

Remark 2.5 (The linear base is the double-negative example)*.* **Vec** fails all the good properties simultaneously. It is not extensive: the comparison $A \amalg B \to A \oplus B$ from the disjoint union of underlying sets to the biproduct misses every genuine linear combination $a + b$, so $\coprod^{\mathbf{Set}} \subsetneq \oplus$. It is not LCC: pullback along a non-mono has no right adjoint, so there is no dependent product Π [8]. Its unit k is a connected *object* but $\mathcal{C}(k, -)$ — the underlying-set functor — does not preserve $\oplus$, so k is *not connected* in the sense of Definition 2.4. And its tiny objects are exactly the finite-dimensional ones, so "tiny" is a genuine restriction, not automatic. Each clause powers one obstruction below.

3. The landscape of bases

The map is worth drawing because the construction it maps keeps being redrawn. The container/polynomial-functor account of a compositional construction is, with some regularity, rediscovered from the implementation side by authors who do not cite — and often do not know of — one another. Hedges' experimental language *Poly* [9], whose types *are* containers and whose functions *are* lenses, is a sharp recent instance: a working language built directly on the Abbott–Altenkirch–Ghani [1] and Gambino–Kock [8] theory, its compiler targeting a graded monad modelling a stack — over the classical base, but arrived at from programming practice rather than from the categorical literature. When a construction is independently reimplemented this often, a census of *where* it has been run, and which of its forms survives which base, stops being pedantry: it is what lets the reimplementers locate one another, and it is what the rest of this section supplies.

Table 1 named the two endpoints of the story: the good base **Set**, where all four approaches coincide, and the doubly-bad base **Vec**, where only the external one survives. Between them lies a whole territory of bases over which some author has actually run the polynomial/container construction, each keeping part of the base structure and losing the rest. Before proving anything we map that territory, because the map is itself an argument for the thesis: every base that has been used sits at a definite point on the two axes, and which approach reaches it is decided by which axis it keeps.

Base $\mathcal{C}$	who ran it	ext.?	LCC?	what the construction reaches
Set, lextensive topos	[1, 8, 13]	✓	✓	all four approaches coincide
finite-limit category	[14]	—	×	$\lhd$-comonoids = internal categories
pullbacks + exp. legs	[16]	—	partial	(2) via distributivity pullbacks
LSCC (quantale, affine)	[15]	×	×	(4) affine $\nabla_f \dashv \boxtimes_f$
any monoidal $\mathcal{V}$	[5]	—	—	(1) $\otimes, \lhd$; $\lhd$-comon. = enr. cats
any fibration q	[13]	—	—	(3) $\mathbf{Cont}(q) = \mathrm{Fam}(q^{\mathrm{op}})$
$\mathbf{Set}^I$ (indexed)	[6]	✓	✓	(1)/(2) composition monoids
Prof (gen. species)	[7]	—	—	cartesian closed bicategory
Vec, R-Mod	this survey	×	×	**(1) only** (T1, T2, T4)
Rel; **Mod**, **Poly**	[12] Ch. 9	?	?	open

TABLE 2. The bases over which the polynomial/container construction has been run, ordered by decreasing base strength. The two centre columns are the discriminating axes of the thesis; "—" marks a generality-of-base result, for which the axes are not the natural descriptor, and "partial" a per-morphism weakening of Π. The middle block (finite-limit category through LSCC) is the weakening tower. **Vec** and R-Mod sit at the doubly-negative corner reached by the external approach alone.

Reading down the table, the bases that keep pullbacks but lose full local cartesian closure form a *totally ordered weakening tower*, each rung smoothing a different seam of the $\Sigma\Pi\Delta$ machinery:

$$\underbrace{\text{LCCC}}_{[8]} \supset \underbrace{\text{pullbacks} + \text{exp. legs}}_{\text{Weber [16]}} \supset \underbrace{\text{subpullbacks (LSCC)}}_{\text{Walker [15]}} \supset \text{spans.}$$

Local cartesian closure (Gambino–Kock) asks pullback to have a right adjoint globally; Weber's *distributivity pullbacks* [16] weaken this to a per-morphism condition — a canonical comparison δ is invertible iff the relevant middle leg is exponentiable — so that composition of polynomials is defined leg-by-leg rather than by a blanket Π; Walker's subpullbacks [15] weaken it further to an affine base-change $\nabla_f \dashv \boxtimes_f$; and at the bottom sit bare spans, with no base-change adjoint at all. The rungs are known individually; what has not been done, and what the survey needs, is to assemble them into one ordered scale and ask where a given base — in particular **Vec** — sits on it. The answer, developed in Section 7, is that **Vec** falls off the bottom: not even the affine rung holds.

The tower refines only one of the two axes — local cartesian closure, which controls the substitution product $\lhd$ and the internal homs. The other axis, extensivity, is orthogonal to it and controls whether the external shape-coproduct is seen correctly, hence fullness of $[\![-]\!]$. This gives the organising fact of the whole landscape.

> *Off the good zone, a base fails to support the classical container theory for exactly one of two independent reasons: loss of extensivity ($\coprod^{\mathbf{Set}} \subsetneq \oplus$, so shapes are no longer recoverable and $[\![-]\!]$ is no longer full) or loss of internal Π (no dependent product, so $\lhd$ and closedness collapse). Toposes lose neither; quantales and affine bases lose only the second, which is why Walker's affine repair reaches them; additive bases such as* **Vec** *and* R-Mod *lose both, which is why only the external approach reaches them.*

This is the two-column heading of Table 2 promoted to the reading of the whole table, and it is the qualitative form of the thesis that the three theorems of Section 4 make quantitative.

Remark 3.1 (Where the two proved theorems sit in Weber's work). The weakening tower places the substitution obstruction and the tensor obstruction in *different* parts of Weber's work, and it is worth separating them to forestall a tempting conflation. The hypothesis of the collapse theorem T4 (Theorem 4.17) — that the left, composition-side positions be tiny (exponentiable) — is exactly Weber's distributivity-δ condition of [16]: δ is invertible iff the middle leg is exponentiable, which under the extension transports to tininess of the left positions and is precisely what makes $\lhd$ well-defined. The hypothesis of the closedness theorem T2 (Theorem 4.10) — that a functor Φ be *familially representable* — is instead a comparison from Weber's *separate* theory of parametric right adjoints and familial functors [17], where a functor is familial iff it is fibrewise a coproduct of representables. These conditions are logically independent: δ constrains the left (composition) positions, Φ the right (tensor) positions and the target, and one can hold while the other fails — a finite-dimensional-left, infinite-dimensional-right datum over **Vec** makes δ invertible while Φ is unrepresentable, and the reverse datum does the opposite. So T4 belongs to the distributivity-pullback rung of the tower, whereas T2 does *not* lie on the tower at all: it lives in the parallel familial-functor theory. This re-filing is a reading of the two Weber papers against the theorems, not a theorem of its own, and we mark it as such.

4. Approach (1): external families

This is the approach that reaches **Vec**, and the one whose obstructions we can measure. We prove the three theorems that quantify the discriminator. All statements are at the level of the *set* of enriched natural transformations, which exists for any closed $\mathcal{C}$ with coproducts; no completeness of $\mathcal{C}$ is needed.

4.1. **The hom formula.** Everything reduces to one enriched computation and one comparison map.

Lemma 4.1 (Enriched co-Yoneda)**.** *For any $\mathcal{C}$-endofunctor G and $P \in \mathcal{C}$,* $\mathrm{Nat}([P,-],G) \cong \mathcal{C}(I,GP)$*, naturally.*

Proof. $[P,-] = \mathcal{C}(P,-)$ is corepresentable; enriched Yoneda evaluates a transformation at id_P, and $\mathcal{C}$-linearity of G on hom-objects recovers it everywhere. Taking points ($\mathcal{C}(I,-)$ of the enriched hom) gives the stated set-level form. □

Lemma 4.2 (Transformations out of a coproduct)**.** $\mathrm{Nat}(\coprod_s F_s, G) \cong \prod_s \mathrm{Nat}(F_s, G)$.

Proof. Coproducts of endofunctors are pointwise, and a transformation out of a pointwise coproduct is a tuple of transformations. □

Combining, for the extensions,

$$\mathrm{Nat}\big([\![S,P]\!],[\![T,Q]\!]\big) \;\cong\; \prod_{s\in S}\; \mathcal{C}\Big(I,\; \textstyle\coprod^{\mathcal{C}}_{t\in T}[Q_t,P_s]\Big). \tag{1}$$

Comparing (1) with the container hom of Definition 2.1, the action of $[\![-]\!]$ on morphisms is, in each shape s, the following canonical map.

Definition 4.3 (The comparison map)**.** For a family $(X_t)_{t\in T}$ in $\mathcal{C}$,

$$\gamma_{(X_t)} \;:\; \textstyle\coprod^{\mathbf{Set}}_t\; \mathcal{C}(I,X_t) \;\longrightarrow\; \mathcal{C}\Big(I,\; \textstyle\coprod^{\mathcal{C}}_t X_t\Big), \qquad (t,\, x\colon I\to X_t) \;\longmapsto\; \mathrm{inj}_t\circ x.$$

This is the comparison testing whether $\mathcal{C}(I,-)$ preserves the coproduct $\coprod_t X_t$. Under $\mathcal{C}(Q_t,P_s) = \mathcal{C}(I,[Q_t,P_s])$, the s-component of $[\![-]\!]$ on morphisms is exactly $\gamma_{([Q_t,P_s])_t}$.

4.2. **T1: fullness is unit-connectedness.**

Theorem 4.4 (Fullness criterion)**.** *Let $\mathcal{C}$ be closed symmetric monoidal with small coproducts.*

(1) *If the unit I is connected then $[\![-]\!]\colon \mathrm{Fam}(\mathcal{C}^{\mathrm{op}}) \to \mathcal{C}\text{-}[\mathcal{C},\mathcal{C}]$ is fully faithful.*
(2) *If $[\![-]\!]$ is full (resp. faithful) then for every object Z and set T the map γ is surjective (resp. injective) for the constant family $(Z)_{t\in T}$; equivalently $\mathcal{C}(I,-)$ preserves (resp. reflects) all copowers $T\cdot Z$. A disconnected unit obstructs fullness.*
(3) *If moreover $\mathcal{C}$ has an object Z_0 with $[-,Z_0]$ essentially surjective onto the objects appearing in the relevant coproducts (e.g. $\mathcal{C} = \mathbf{Vec}_{\mathrm{fd}}$, $Z_0 = k$, $[-,k] = (-)^*$), then $[\![-]\!]$ is fully faithful iff I is connected.*

Proof. (1) If $\mathcal{C}(I,-)$ preserves coproducts, $\gamma_{(X_t)}$ is a bijection for *every* family, so by (1) every shape-component of the action is a bijection.

(2) Take source $(\{s\},Z)$ and target $(T,(I)_{t\in T})$, so $[I,Z]\cong Z$ and the family is constant Z; the s-component of the action is $\gamma_{(Z)_{t\in T}}\colon T\cdot\mathcal{C}(I,Z)\to\mathcal{C}(I,T\cdot Z)$. Full forces surjectivity for all T,Z (i.e. preservation of copowers), faithful forces injectivity.

(3) With Z_0 as stated, $\{[Q,Z_0] : Q\in\mathcal{C}\}$ is, up to isomorphism, all objects, so the families $([Q_t,Z_0])_t$ range over all set-indexed families; fullness plus faithfulness then give that $\mathcal{C}(I,-)$ preserves all small coproducts, i.e. I is connected. With (1) this is the stated equivalence. □

The point of Theorem 4.4 is that it separates two properties the **Set** case fuses.

Corollary 4.5 (**Set**: recovers Abbott–Altenkirch–Ghani)**.** *For $\mathcal{C} = \mathbf{Set}$, $I = 1$ and $\mathbf{Set}(1,-) = \mathrm{id}$ preserves coproducts, so $[\![-]\!]$ is fully faithful — the representation theorem of [1]. The hidden hypothesis is that the terminal object of* **Set** *is connected.*

Corollary 4.6 (**Set** $\times$ **Set**: extensive but not full)**.** *$\mathbf{Set}\times\mathbf{Set}$ is (l)extensive, yet its unit $(1,1)$ is disconnected:*

$$\mathcal{C}\big((1,1),(X,Y)+(X',Y')\big)=(X+X')\times(Y+Y')\ \neq\ X\times Y+X'\times Y',$$

the comparison missing the cross terms (x,y'). By Theorem 4.4(2) $[\![-]\!]$ is not full: with source $(\{s\},(1,1))$ and target $(\{t_1,t_2\},((1,1),(1,1)))$ there are 2 container morphisms but $\mathcal{C}((1,1),(2,2))=4$ natural transformations, the two "crossed" transformations unrealised.

Corollary 4.7 (**Vec**: the linear collapse)**.** *For $\mathcal{C}=\mathbf{Vec}$, $I=k$, the functor $\mathbf{Vec}(k,-)$ is the underlying-set functor, which does not preserve $\oplus$: the comparison $\coprod_t^{\mathbf{Set}} X_t\to\bigoplus_t X_t$ is neither surjective (it misses cross-summand sums) nor injective (it collapses zeros). By Theorem 4.4(2,3) with $Z_0=k$, $[\![-]\!]$ is neither full nor faithful over **Vec**, and the failure is exactly the disconnectedness of k. This is the $\coprod^{\mathbf{Set}}\subsetneq\oplus$ of the non-extensivity of **Vec**, now placed as one instance of the general criterion.*

Theorem 4.8 (The corrected slogan)**.** *Extensivity of $\mathcal{C}$ is neither necessary nor sufficient for $[\![-]\!]$ to be full. The controlling invariant is connectedness of the monoidal unit. (**Set** $\times$ **Set** is extensive but not full; **Set** is full not because it is extensive but because 1 is connected.)*

Remark 4.9 (Why the folklore said "extensivity")*.* Two distinct theorems have been conflated: (a) full faithfulness of the fixed extension $[\![-]\!]$ (this section: I connected), and (b) Diers' reconstruction [4] of (S,P) from an abstract familially-representable $F\colon\mathcal{C}\to\mathbf{Set}$ as $S=\pi_0(\mathrm{el}\,F)$, which genuinely uses extensivity of the *codomain* **Set**. For $\mathcal{C}=\mathbf{Set}$ base and codomain coincide, so (a) and (b) fuse and "extensivity" reads off both. Off **Set** they separate: (a) is governed by the unit of the enrichment base, (b) by extensivity of the value category. Similarly this is no contradiction with the fully internal polynomial functors of [8]: those never form an external set-coproduct of internal homs, so their representation theorem is untouched. The external-shape calculus is a **Set**-enriched phenomenon unless the base's unit is connected.

4.3. **T2: closedness of the Dirichlet tensor.** We ask when $(-)\otimes(T,Q)$ has a right adjoint inside $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$.

Theorem 4.10 (Closedness = familial representability)**.** *The following are equivalent.*

1. *$\otimes$ is closed: $(-)\otimes(T,Q)$ has a right adjoint for every (T,Q).*
2. *For all families $(Q_t)_{t\in T}$, $(M_r)_{r\in R}$ the functor $\Phi(Z)=\prod_t\coprod_r\mathcal{C}(M_r,Z\otimes Q_t)\colon\mathcal{C}\to\mathbf{Set}$ is familially representable, i.e. $\Phi\cong\coprod_u\mathcal{C}(N_u,-)$ for a small family (N_u).*
3. *For all families the atom $\Theta_{A,Q}(Z)=\prod_t\mathcal{C}(A_t,Z\otimes Q_t)$ is familially representable.*

When they hold, $[(T,Q)\Rightarrow(R,M)]=(U,(N_u))$.

Proof. (1) $\Leftrightarrow$ (2): a right adjoint at $(T,Q),(R,M)$ is an object representing $G:=\mathrm{Fam}(-\otimes(T,Q),(R,M))\colon\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}\to\mathbf{Set}$. Since $\otimes$ distributes over the shape-coproduct and hom out of a coproduct is a product, G sends coproducts to products, and on the generators $G(\langle Z\rangle)=\prod_t\coprod_r\mathcal{C}(M_r,Z\otimes Q_t)=\Phi(Z)$. By the universal property of the free coproduct completion (Definition 2.1), such a G is representable iff its restriction to the generators is, i.e. iff Φ is familially representable; and the representing family is the internal hom.

(2) $\Leftrightarrow$ (3): for sets, $\prod_t\coprod_r X_{t,r}=\coprod_{\rho\colon T\to R}\prod_t X_{t,\rho(t)}$ (no hypothesis on $\mathcal{C}$), whence $\Phi\cong\coprod_\rho\Theta_{M\circ\rho,Q}$ (the identity (2) below). A coproduct of familially representable functors is familially representable, giving (3) $\Rightarrow$ (2); and each corepresentable is connected in $[\mathcal{C},\mathbf{Set}]$, so $\Theta_{A,Q}$ appears as the $\rho=\mathrm{id}$ summand of a familially-representable Φ and is itself familially representable, giving (2) $\Rightarrow$ (3). $\square$

The set-distributivity used above,

$$\Phi(Z)\ \cong\ \coprod_{\rho\colon T\to R}\ \Theta_{M\circ\rho,Q}(Z),\qquad \Theta_{M\circ\rho,Q}(Z)=\prod_t\mathcal{C}(M_{\rho(t)},Z\otimes Q_t),\tag{2}$$

is the engine of both regimes in which $\otimes$ is closed.

Theorem 4.11 (Two sufficient regimes)**.** (1) (Cartesian.) *If $\otimes = \times$ (so $\mathcal{C}$ is cartesian closed), $\otimes$ is closed. The projection split $\mathcal{C}(A, Z \times Q) \cong \mathcal{C}(A, Z) \times \mathcal{C}(A, Q)$ makes each $\Theta_{A,Q}$ a copower of the corepresentable at $\coprod_t A_t$, and*

$$[(T, Q) \Rightarrow (R, M)] = \Big(\textstyle\coprod_{\rho\colon T \to R} \prod_t \mathcal{C}(M_{\rho(t)}, Q_t),\ \big(\coprod_t M_{\rho(t)}\big)_\rho\Big),$$

the classical Dirichlet internal hom of **Poly** *([12], Ex. 4.78) over* **Set**.

(2) (Rigid.) *If every Q_t is dualizable and $\mathcal{C}$ has the coproducts $\coprod_t A_t \otimes Q_t^*$, then $\otimes$ is closed, with*

$$[(T, Q) \Rightarrow (R, M)] = \Big(R^T,\ (N_\rho)_{\rho\colon T \to R}\Big), \qquad N_\rho = \textstyle\coprod_t M_{\rho(t)} \otimes Q_t^*.$$

Here $Z \otimes Q_t \cong [Q_t^*, Z]$ gives $\mathcal{C}(A_t, Z \otimes Q_t) \cong \mathcal{C}(A_t \otimes Q_t^*, Z)$, so $\Theta_{A,Q}$ is a single corepresentable at $\coprod_t A_t \otimes Q_t^*$.

Over **Vec** neither the projection split nor the diagonal Δ is available, so the only route to representability is the rigid one — and it is also *necessary*.

Lemma 4.12 (Single-factor representability over **Vec**)**.** *For $A, Q \in$ **Vec** the functor $\Theta(Z) =$ **Vec**$(A, Z \otimes Q)$ is familially representable iff Q is finite-dimensional, in which case it is corepresentable at $A \otimes Q^*$.*

Proof. ($\Leftarrow$) is Theorem 4.11(2) with a single t. ($\Rightarrow$): take $A = k$, so $\Theta(Z) = |Z \otimes Q|$ (underlying set), and $\Theta(0) = 1$ forces a familial representation to be a single corepresentable **Vec**$(N, -)$. If $\dim Q = \infty$, write $Q = k^{(I)}$; then $\Theta(Z)$ is the finitely-supported functions $I \to Z$, which fails to preserve an infinite product $\prod_J k$: the element $(e_{i_j})_{j \in J}$ with the i_j pairwise distinct lies in $\prod_J \Theta(k)$ but has no finite-support preimage in $\Theta(\prod_J k)$. So Θ is not corepresentable, hence not familially representable. $\square$

Theorem 4.13 (The linear closedness dichotomy)**.** (1) *On* $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *(finite shape-sets, finite-dimensional positions), $\otimes$ is closed, with internal hom $(R^T, (\bigoplus_t M_{\rho(t)} \otimes Q_t^*)_\rho)$, which stays finite-dimensional.*

(2) *On* $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *(arbitrary shape-sets), $\otimes$ is not closed: for $T = \mathbb{N}$, all $Q_t = k$, $R = \{r\}$, $M_r = k$, one computes $\Phi(Z) = |Z|^{\mathbb{N}} = \mathbf{Vec}(k^{(\mathbb{N})}, Z)$, whose representing object $k^{(\mathbb{N})}$ is infinite-dimensional, so no object of* $\mathbf{Vec}_{\mathrm{fd}}$ *represents it.*

(3) *On* $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ *(infinite-dimensional positions), $\otimes$ is not closed: for a single $Q = k^{(\mathbb{N})}$, $\Phi(Z) = |Z \otimes Q|$ is not familially representable by Lemma 4.12.*

Remark 4.14 (The named obstruction, and its independence from T1). Closedness needs the hom position $N_\rho = \coprod_t M_{\rho(t)} \otimes Q_t^*$ to (i) *make sense* — each Q_t dualizable, forcing finite dimension over a linear base — and (ii) *exist* — the coproduct present in $\mathcal{C}$. Over **Vec** these pull apart the moment T is infinite: (i) pins positions to $\mathbf{Vec}_{\mathrm{fd}}$, (ii) needs a colimit escaping $\mathbf{Vec}_{\mathrm{fd}}$. The single load-bearing conjunct is thus *dualizable-and-summable*, satisfiable over **Vec** exactly on the finite/fd corner. This is a different seam from T1: T1 fullness is the failure of $\mathcal{C}(I, -)$ to preserve $\oplus$ (a property of the unit), whereas T2 is a cocompleteness failure over $\mathbf{Vec}_{\mathrm{fd}}$ and a dualizability failure over full **Vec**. Cartesian bases dodge the collision because the diagonal Δ does the work of duals at no cost — which is why **Set** $\times$ **Set**, though not full (Corollary 4.6), still has $\otimes$ closed. Fullness and $\otimes$-closedness are independent axes.

4.4. **T4: left-closedness of the substitution product.** The substitution product $\lhd$ is non-symmetric, so it has two potential internal homs. Its right internal hom — the right adjoint to $(T, Q) \lhd (-)$ — is the $\lhd$-*coclosure*, which exists over **Set** and gives directed containers ([12], Prop. 6.57). We study the other one: the *left* internal hom, right adjoint to $(-) \lhd (T, Q)$, the $\lhd$-*closure*. Over **Cont** $= \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}})$ it is obstructed.

Proposition 4.15 (The **Set** obstruction)**.** *Over* **Set***,* $(-) \lhd q$ *has no right adjoint. Computing the extension of the generator,* $[\![\langle Z\rangle \lhd q]\!]X = [Z, \coprod_t [Q_t, X]]$*, and the exponential distributes over the coproduct,*

$$[Z, \textstyle\coprod_t B_t] = \coprod_{\tau\colon Z\to T} \prod_{d\in Z} B_{\tau(d)}, \tag{3}$$

so $\langle Z\rangle \lhd q$ *has shape-set* T^Z*. By the universal property of the free coproduct completion (Definition 2.1), a right adjoint would be a container representing* $(-) \lhd q$ *generator by generator; but the* $|T|^{|Z|}$*-fold shape-set makes its shape-count grow double-exponentially (*$|T_R([n])| \sim n^{2^n}$ *in the free case), whereas every container has a* polynomial *extension. No container carries such data, so the right adjoint does not exist for* $|T| \geq 2$*.*

The obstruction (3) is created entirely by $[Z, -]$ failing to preserve $\coprod_t$ — the distributive law of an extensive, cartesian closed base. Tininess removes it.

Proposition 4.16 (Collapse $\lhd = \otimes$)**.** *If every position object of* $p = (S, P)$ *is tiny, then* $p \lhd q = p \otimes q = (S \times T, (P_s \otimes Q_t))$ *for every* $q = (T, Q)$*.*

Proof. Tininess replaces (3) by genuine coproduct-preservation:

$$[\![p \lhd q]\!]X = \textstyle\coprod_s [P_s, \coprod_t [Q_t, X]] = \coprod_{s,t} [P_s, [Q_t, X]] = \coprod_{s,t} [P_s \otimes Q_t, X] = [\![p \otimes q]\!]X,$$

using tininess of P_s in the middle step and the tensor–hom adjunction at the end. □

Over $\mathbf{Vec}_{\mathrm{fd}}$ all objects are dualizable, hence tiny, so Proposition 4.16 makes $\lhd$ symmetric and identifies its left internal hom with the Dirichlet hom of Theorem 4.13.

Theorem 4.17 ($\lhd$-left-closedness over the linear base)**.** *With* $\lhd$ *as above:*

1. $\mathrm{Fam}_{\mathrm{fin}}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *is* $\lhd$*-left-closed:* $\lhd = \otimes$ *by Proposition 4.16, and* $\otimes$ *is closed there by Theorem 4.13(1). The left internal hom is*

$$[(T, Q) \lhd (R, M)] = \Big(R^T,\ \big(\textstyle\coprod_t M_{\rho(t)} \otimes Q_t^*\big)_{\rho\colon T\to R}\Big).$$

2. $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *is not* $\lhd$*-left-closed (infinite shape-sets break summability, as in Theorem 4.13(2)).*
3. $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ *is not* $\lhd$*-left-closed (infinite-dimensional positions are not tiny, so the collapse itself fails, as in Theorem 4.13(3)).*

Theorem 4.18 (Extensivity is opposed to $\lhd$-left-closedness)**.** *The* **Set** *obstruction (Proposition 4.15) and the* $\mathbf{Vec}_{\mathrm{fd}}$ *repair (Theorem 4.17) are two valuations of the one fact — the distributive law (3) — with opposite sign.*

- *Over an extensive, cartesian closed base,* $[Z, -]$ *explodes* $\coprod_t$ *into* T^Z *branches:* $\lhd \neq \otimes$*,* $\lhd$ *is genuinely non-symmetric, and its closure is non-polynomial (absent). Extensivity* is *the obstruction.*
- *Over a linear base,* $[Z, -]$ *is additive and preserves* $\coprod_t$*:* $\lhd$ *collapses onto the closed, symmetric* $\otimes$*. Non-extensivity* is *the repair.*

This inverts T1 and T2, where non-extensivity was the obstruction. The reason: T1/T2 measure whether the external **Set***-coproduct on shapes is seen correctly by the base (non-extensivity breaks that), while* $\lhd$*-closedness needs the base's internal hom on positions* not *to branch against coproducts — which is exactly additivity. The same* $\coprod^{\mathbf{Set}} \subsetneq \oplus$ *seam that costs fullness buys* $\lhd$*-closedness.*

base $\mathcal{C}$	extensive?	$\lhd$ vs $\otimes$	$\lhd$ symmetric?	$\otimes$-closed?	$\lhd$-left-closed?
Set (toposes)	✓	$\lhd \neq \otimes$	no	yes	**no**
$\mathbf{Vec}_{\mathrm{fd}}$ (fin. corner)	×	$\lhd = \otimes$	yes	yes	**yes**

TABLE 3. The inversion of Theorem 4.18: extensivity, the villain of T1/T2, is the hero of the $\lhd$-closure obstruction.

5. APPROACH (2): INDEXED AND DEPENDENT-POLYNOMIAL FUNCTORS

The second approach keeps the shapes *internal*. A polynomial in the sense of Gambino and Kock [8] is a diagram

$$I \xleftarrow{s} P \xrightarrow{f} S \xrightarrow{t} O$$

in a base $\mathcal{C}$, and its extension is the composite of slice functors

$$\mathcal{C}/I \xrightarrow{\Delta_s} \mathcal{C}/P \xrightarrow{\Pi_f} \mathcal{C}/S \xrightarrow{\Sigma_t} \mathcal{C}/O, \tag{4}$$

where $\Delta_s = s^*$ is pullback (reindexing), Π_f the dependent product (right adjoint to f^*), and Σ_t the dependent sum (left adjoint to t^*). Setting $I = O = 1$ collapses the outer slices to $\mathcal{C}$ itself and (4) becomes an endofunctor $\mathcal{C} \to \mathcal{C}$; a "container in $\mathcal{C}$" in this sense is then an internal map $P \to S$, with extension $X \mapsto \Sigma_S \Pi_P X$. Over **Set** this is $\coprod_{s \in S} X^{P_s}$ — the classical container extension, and the reason the two approaches coincide there.

Remark 5.1 (The load-bearing hypothesis). The functor Π_f exists precisely when $\mathcal{C}$ is locally cartesian closed. This is not a convenience: without it the middle arrow of (4) is undefined, and the approach cannot even *form* its semantics — there is no endofunctor to speak of. Over an extensive LCC base the resulting category of polynomials is itself extensive and reproduces the whole classical picture, which is Neil Ghani's expectation and is correct for **Set**, any topos, and any LCC pretopos.

Proposition 5.2 (The approach cannot reach **Vec**)**.** **Vec** *is not locally cartesian closed (Remark 2.5), so the dependent product* Π_f *in* (4) *does not exist. Approach (2) therefore cannot form its semantics over* **Vec***: there is no* $\Sigma\Pi\Delta$ *polynomial functor associated to an internal map of vector spaces. This is a* negative *result — not that the polynomial functors over* **Vec** *are badly behaved, but that they are not defined.*

This is the first half of the discriminator. Where approach (1) sees $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ directly — its external set-coproduct never asks $\mathcal{C}$ for an internal Π — approach (2) is blocked at the definition. De Pascalis, Uustalu and Veltrì [6] give the composition-monoid refinement of the single-index corner of this approach (a common index, base **Set**, dropping the cartesianness of [8]); their indexed containers are of the same lineage as (2), still over **Set**, and we return to them in Section 9 as the target of the sharpest open problem.

6. APPROACH (3): THE FIBRATIONAL REFEREE, AND THE LOGIC OF CONTAINERS

The third approach begins as a referee. Present the position data as a fibration $p\colon \mathcal{E} \to \mathbb{B}$ with shapes in the base $\mathbb{B}$ and positions in the fibres, and give container semantics either by comprehension down to the base or directly in the total category. The two earlier approaches are then two instances of one scheme:

- approach (1) is the *family fibration* $\mathrm{Fam}(\mathcal{C}) \to \mathbf{Set}$: shapes index an external set, fibres are objects of $\mathcal{C}$;

- approach (2) is the *codomain fibration* $\mathrm{cod}\colon \mathcal{C}^{\to} \to \mathcal{C}$ of an LCC base: shapes and positions are both internal, and Π is the fibrewise right adjoint that the codomain fibration of an LCC category carries.

Remark 6.1 (Where the fibrations diverge). The question "which approach" becomes "which fibration models your positions." Over an extensive LCC base the family and codomain fibrations agree, and the choice is presentational. Over a base that fails one property they genuinely diverge: **Vec** has a family fibration $\mathrm{Fam}(\mathbf{Vec}) \to \mathbf{Set}$ (so approach (1) is available) but no codomain fibration with dependent products (so approach (2)'s Π is absent). The fibrational view thus reads the discriminator as a statement about which fibration exists.

But the fibrational approach is more than a referee. Applying the container construction $\mathbf{Cont}(-) := \mathrm{Fam}((-)^{\mathrm{op}})$ to the codomain fibration of **Set** itself produces a fibration *of containers over containers* — a bifibration whose fibres are proof-relevant predicates on positions. This is the "logic of containers," and it is where the predicate structure that approaches (1) and (2) merely presuppose becomes an explicit calculus. We develop it now; it is the positive content of the third approach and, at the level of the quantifiers, it recovers the All and Exists predicate liftings on positions — the fibred dependent sum and product.

6.1. Fam **preserves fibrations.** The construction rests on a single structural lemma. Recall $\mathrm{Fam}(\mathcal{D})$ has objects $(I, (X_i)_{i\in I})$ with $I \in \mathbf{Set}$, and morphisms $(u, (f_i))\colon (I, (X_i)) \to (J, (Y_j))$ a function $u\colon I \to J$ with $f_i\colon X_i \to Y_{u(i)}$; for $p\colon \mathcal{E} \to \mathbb{B}$ set $\mathrm{Fam}(p)(I, (X_i)) = (I, (pX_i))$.

Lemma 6.2 (Componentwise cartesianness). *Let $p\colon \mathcal{E} \to \mathbb{B}$ be any functor. A morphism $(u, (f_i))$ of $\mathrm{Fam}(\mathcal{E})$ is $\mathrm{Fam}(p)$-cartesian if and only if each f_i is p-cartesian. Consequently, if p is a fibration (resp. opfibration, bifibration) then so is $\mathrm{Fam}(p)$, with (op)reindexing computed componentwise.*

Proof sketch. Because $\mathrm{Fam}(\mathcal{E})$ is the free coproduct completion, a morphism carries no cross-index datum beyond the reindexing function u; a factorisation problem for $(u, (f_i))$ therefore decomposes into one independent factorisation problem per index i, each solved uniquely exactly when f_i is p-cartesian. The cartesian lift of a base morphism $(u, (\psi_i))$ into $(J, (Y_j))$ is $(u, (\chi_i))$, where χ_i is the p-cartesian lift of ψ_i and the position object is $X_i := \psi_i^* Y_{u(i)}$. The opcartesian statement is the same argument reversed. The verification is a routine diagram chase; the fact is folklore [11, Ch. 1], [10]. $\square$

Theorem 6.3 (The bifibration of containers). *The codomain functor* $\mathrm{cod}\colon \mathbf{Set}^{\to} \to \mathbf{Set}$ *is a bifibration (opreindexing Σ_β is post-composition; reindexing β^* is pullback, available because* **Set** *has pullbacks). Hence* $\mathrm{cod}^{\mathrm{op}}\colon \mathbf{Set}^{\to\mathrm{op}} \to \mathbf{Set}^{\mathrm{op}}$ *is a bifibration, and by Lemma 6.2*

$$\mathbf{Cont}(\mathrm{cod}) \;:=\; \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}}) \;:\; \mathbf{Cont}(\mathbf{Set}^{\to}) \longrightarrow \mathbf{Cont}(\mathbf{Set})$$

is a bifibration. Its fibre over a container $(S, (P_s)_{s\in S})$ is

$$\textstyle\prod_{s\in S} (\mathbf{Set}/P_s)^{\mathrm{op}},$$

the fibrewise opposite *of the slices: an object assigns to each shape s and each position $b \in P_s$ a witness set, i.e. a* proof-relevant predicate on positions*.*

Proof sketch. cod is Streicher's fundamental fibration; the op of a bifibration is a bifibration, so $\mathrm{cod}^{\mathrm{op}}$ is one and Lemma 6.2 lifts it through Fam. An object of $\mathbf{Cont}(\mathbf{Set}^{\to})$ is $(S, (f_s\colon A_s \to B_s))$ and $\mathbf{Cont}(\mathrm{cod})$ returns $(S, (B_s))$; a vertical morphism over $\mathrm{id}_{(S,(P_s))}$ unwinds to a $\mathbf{Set}/P_s$-morphism in the *backward* direction for each s, whence the fibrewise op. This is von Glehn's presentation $\mathbf{Cont} = \int_{\mathbf{Set}} \mathrm{cod}^{\mathrm{op}}$ [13]. $\square$

The fibrewise op is not an accident of bookkeeping: it is forced by the contravariance of container morphisms on positions, and it is what makes the resulting logic *dual* to the familiar one.

6.2. **The quantifiers are the All/Exists liftings.** Fix a container $(S,(P_s))$. Between the trivial predicate $(S,(1)_s)$ and $(S,(P_s))$ there is one canonical base morphism, the *collapse* $\eta\colon (S,(1)) \to (S,(P_s))$ carrying the backward position map $!\colon P_s \to 1$ (the reverse direction would require a choice of point and is not canonical). Because **Set** is locally cartesian closed, collapse along $!\colon P_s \to 1$ gives the standard adjoint string in each slice,

$$\Sigma_! \dashv !^* \dashv \Pi_! \qquad (\,\exists \dashv \text{weakening} \dashv \forall\,),$$

with $\Sigma_!(f\colon A \to P_s) = A$ the total space, $!^*Z = (P_s \times Z \to P_s)$ the constant family, and $\Pi_!(f) = \prod_b f^{-1}(b)$ the set of sections. Transporting this string through the fibrewise op and Fam (an adjunction $L \dashv R$ becomes $R^{\mathrm{op}} \dashv L^{\mathrm{op}}$) yields the fibred quantifiers of $\mathbf{Cont}(\mathrm{cod})$ along η.

Theorem 6.4 (Quantifiers = liftings). *Along the collapse η, the bifibration $\mathbf{Cont}(\mathrm{cod})$ carries a two-sided adjoint string*

$$\mathsf{All} \dashv \Delta_c \dashv \mathsf{Exists},$$

in which the cartesian reindexing $\eta^* = (\Sigma_!)^{\mathrm{op}}$ is Exists *(the total space), the opreindexing $\eta_! = (!^*)^{\mathrm{op}} = \Delta_c$ is weakening, and $\mathsf{All} = (\Pi_!)^{\mathrm{op}}$ is the fibrewise product of witnesses. These are the* All *and* Exists *predicate liftings on positions:* Exists *is the dependent sum Σ_b over positions,* All *the dependent product $\prod_b$.*

Remark 6.5 (The reindexing is $(\Sigma_\rho)^{\mathrm{op}}$, not ρ^*). It is tempting to guess that reindexing along a container morphism $(u,(\rho_s))$ is pullback ρ_s^*. It is not: by Lemma 6.2 the cartesian lift is computed componentwise from the $\mathrm{cod}^{\mathrm{op}}$-cartesian lift, which by op-duality is the cod-*co*cartesian lift Σ_{ρ_s} dualised. So the cartesian reindexing of $\mathbf{Cont}(\mathrm{cod})$ is post-composition $(\Sigma_\rho)^{\mathrm{op}}$, sending a predicate $g\colon C \to P'_{u(s)}$ to $\rho_s \circ g\colon C \to P_s$. Pullback ρ_s^* reappears, but as the *op*reindexing (weakening) of the opfibration — the left adjoint, not the cartesian direction.

6.3. **The dualisation: containers are the fibrewise-opposite hyperdoctrine.** The adjoint string of Theorem 6.4 is the fibrewise opposite of the **Set** string $\Sigma_! \dashv !^* \dashv \Pi_!$. This one fact reorganises the whole predicate logic.

Theorem 6.6 (Dualisation). *Relative to the common weakening Δ_c, the container logic's existential (left adjoint) is* All*, built from* **Set***'s dependent* product Π*, and its universal (right adjoint) is* Exists*, built from* **Set***'s dependent* sum Σ*: container-$\exists$ is* **Set**-$\forall$ *and container-$\forall$ is* **Set**-$\exists$*. In each fibre $(\mathbf{Set}/P_s)^{\mathrm{op}}$ the fibre product (container $\wedge$) is disjoint union of witness sets, the fibre coproduct (container $\vee$) is the fibrewise product of witnesses, $\top$ is the empty-witness predicate $\emptyset \to P_s$, and $\bot$ is the full predicate* $\mathrm{id}\colon P_s \to P_s$*. Hence the hyperdoctrine of containers is the* fibrewise opposite *of the standard* **Set***-family hyperdoctrine, with $\exists \leftrightarrow \forall$, $\wedge \leftrightarrow \vee$, $\top \leftrightarrow \bot$ all exchanged, and each fibre is a* co-topos *whose internal logic is co-intuitionistic (subtractive).*

The names All and Exists track the **Set** operation used to build the lifting (Π resp. Σ); Theorem 6.6 shows their *logical role* in the container hyperdoctrine is the reverse. The subtractive flavour is genuine only at the proof-relevant level: the crude Boolean truncation $\mathrm{Sub}(P_s) = 2^{P_s}$ is self-dual and hides it, whereas the witness-level subobject fibration $\mathrm{Sub}(\mathbf{Set}^{\to}) \to \mathbf{Set}$ retains it. Beck–Chevalley and Frobenius reciprocity hold, dualised: for squares in $\mathbf{Cont}(\mathbf{Set})$ that are componentwise the op-image of **Set** pullback squares, Exists and All commute with substitution, and All satisfies Frobenius with $\wedge$ replaced by the fibrewise coproduct $\vee$ (co-Frobenius).

Remark 6.7 (Scope of the logic). Two honest limitations. First, Theorems 6.4–6.6 describe the *position-level* (fibrewise) logic only; the full first-order logic over $\mathbf{Cont}(\mathbf{Set})$ also needs quantification over shapes, which is the Fam coproduct-completion structure, and we do not verify here that the combined shape-and-position hyperdoctrine satisfies Beck–Chevalley and Frobenius jointly. Second, the Beck–Chevalley squares are identified only as the op-image of **Set** pullback squares; an intrinsic characterisation in terms of container pullbacks is left open. Neither gap affects Theorems 6.3–6.6 as stated.

Remark 6.8 (Enriched Yoneda and the set-level T1)*.* Presenting each object of $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ as a presheaf $\mathcal{C} \to \mathbf{Set}$ invites an *enriched*-Yoneda statement of full faithfulness. Our proof of Theorem 4.4 is deliberately set-level: it factors through the single comparison map γ (Definition 4.3) and needs no enrichment of the ambient functor category. The enriched statement — that $[\![-]\!]$ is a fully faithful $\mathcal{C}$-functor — is the same content repackaged: enriched full faithfulness is exactly the family of isomorphisms (1), whose points are the γ-bijections. The set-level route is preferable here because the invariant it isolates (connectedness of the unit) is a property of $\mathcal{C}(I, -)$, not of the enrichment, and this is what makes the $\mathbf{Set} \times \mathbf{Set}$ counterexample legible.

For the discriminator, approach (3) plays two roles at once. As referee it reads "which approach" as "which fibration models your positions," and the fibrations part company precisely at the non-extensive, non-LCC bases (Remark 6.1). As a construction in its own right over **Set** it supplies the predicate calculus — the bifibration $\mathbf{Cont}(\mathrm{cod})$ and its dualised quantifiers — that names the logical structure the container of positions carries, and identifies it with von Glehn's fibrewise op and the $\mathsf{All}/\mathsf{Exists}$ liftings.

7. Approach (4): the subcartesian weakening

One might hope to rescue approach (2) over **Vec** by weakening the requirement that costs it Π. Walker's *locally subcartesian closed* (LSCC) categories [15] do exactly this, along a lineage (Weber distributivity pullbacks, Street protocalibrations) independent of Gambino–Kock. The idea is to equip a category having pullbacks with a coherent choice of *subpullbacks* — chosen subobjects of the genuine pullbacks — from which one builds *affine* base-change functors ∇_f whose right adjoints $\boxtimes_f$ (the "dependent subproduct") can exist even when the top adjoint $\Delta_f \dashv \Pi_f$ of the LCC chain does not. Each slice then carries a subcartesian tensor $f \otimes_X g = \Sigma_g \nabla_g(f)$ that is right-closed without being cartesian closed; the motivating example is the Lawvere quantale, with the additive, non-cartesian slice tensor $A + B - X$.

This is a genuine fourth weakening, and its additive slice tensor is structurally close to the $\otimes/\oplus$ shape of **Vec**. So it sharpens the discriminator: does **Vec**, though it fails full LCC, carry a coherent subpullback structure making it LSCC?

Remark 7.1 (Verdict: **Vec** is not locally subcartesian closed)*.* Reading Walker's definitions against **Vec** gives a negative answer, on three load-bearing points [15, Def. 4.0.1, Rem. 4.0.2, Rem. 4.0.4, fn. 15].

1. *Pullbacks are mandatory* (Rem. 4.0.4): the construction uses Dubuc's adjoint-triangle theorem, which needs each Σ_f to have a right adjoint, so $\Sigma_f \dashv \Delta_f$ always holds and only the top adjoint is weakened. The chain does not fully fail; just its top.
2. *Closure demands genuine right adjoints* $\boxtimes_p$ (Def. 4.0.1): the slice tensor must be *right closed*, i.e. each affine base-change must have an actual right adjoint. The "unique-if-exists" relaxation applies only to the subpullback (the ∇ side), not to the closure step, so it gives no back door.
3. *The nearest cousin is filed as obstructed* (Rem. 4.0.2, fn. 15): the category of *affine spaces* is "somewhat close to working" — it has a canonical $\boxtimes_{X\to 1}$ because slice morphisms are closed under affine combinations — but general $\boxtimes_p$ fails, the obstruction being *cross-fiber combinations*. Walker "mostly ignores such cases as they lack the desired polynomial structure."

The cross-fiber obstruction of (3) is exactly the $\coprod^{\mathbf{Set}} \subsetneq \oplus$ biproduct collapse of **Vec** in disguise: linear combinations that cross fibres are precisely what the biproduct adds and the coproduct-of-underlying-sets lacks. Walker's polynomial biequivalence (Thm. 5.2.8) is a recognition theorem *assuming* local subcartesian closure, not a criterion certifying it for a given non-LCC base, so it offers no route around the obstruction.

Remark 7.2 (Status of the verdict). Remark 7.1 is a reading of Walker's preprint checked against the structure of **Vec**, not a theorem with a proof of our own; we state it as an understanding result. Its role in the survey is to close the last escape hatch: **Vec** fails not only extensivity and local cartesian closure but even Walker's weaker affine substitute. The discriminator gap is therefore not "LCC or not" but the deeper "no affine base-change survives at all."

8. Synthesis: the discriminator

We can now state the classification the three (four) approaches obey.

Theorem 8.1 (The two-axis discriminator). *Let $\mathcal{C}$ be a closed symmetric monoidal base.*

(1) *If $\mathcal{C}$ is extensive and locally cartesian closed, all four approaches agree: the external family, indexed-polynomial, family-fibration and codomain-fibration constructions coincide, and one recovers the classical convergence — containers $\simeq$ directed containers $\simeq$* **Poly***-comonoids $\simeq$ (small) categories.*

(2) *If $\mathcal{C}$ fails local cartesian closure, approach (2) cannot form its $\Sigma\Pi\Delta$ semantics (Proposition 5.2); if $\mathcal{C}$ additionally fails Walker's subpullback closure, approach (4) cannot either (Remark 7.1).*

(3) *The external approach (1) reaches any closed $\mathcal{C}$ with coproducts, using external set-indexing and never asking $\mathcal{C}$ for an internal Π or a well-behaved $\coprod$. Its obstructions are not to* existence *but to* good behaviour*, and they are measured exactly:*
   - *full faithfulness fails iff the unit is disconnected (Theorem 4.4);*
   - *$\otimes$-closedness fails off the cartesian bases and the finite-fd linear corner (Theorem 4.13);*
   - *$\lhd$-left-closedness holds only where $\lhd$ collapses onto $\otimes$, i.e. on the same finite-fd corner, and is obstructed on every extensive base (Theorems 4.17, 4.18).*

The base **Vec** *realises the corner that has neither property: only approach (1) reaches* $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$*, and the three obstructions of (3) are the quantitative face of the two failed hypotheses.*

The classification is genuinely two-dimensional. Extensivity and local cartesian closure are independent (Table 1 has a base failing each), and they act on the container structures with *opposite* sign on the two tensors: non-extensivity breaks fullness and $\otimes$-closedness (T1, T2) but repairs $\lhd$-closedness (T4), because the first two ask the base to see an external coproduct correctly while the third asks its internal hom not to branch against coproducts. A single "goodness" scale cannot express this; the two axes can. $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$ is the object that makes the two-dimensionality visible, because it is the corner where both axes are at their extreme.

9. Conclusion and the sharpest open problem

Four constructions answer to "a container in a category," and they diverge along two independent properties of the base: extensivity and local cartesian closure. When the base has both, the four agree and the classical theory is recovered. When the base has neither — the linear base **Vec** — only the external-family construction survives, and its survival is exactly measured by three dichotomies: full faithfulness tracks connectedness of the unit, $\otimes$-closedness tracks a dualizable-and-summable condition met only on the finite-dimensional corner, and $\lhd$-left-closedness holds precisely where linearity collapses the substitution product onto the Dirichlet tensor. The last inverts the first two: the non-extensivity that costs fullness and $\otimes$-closedness is what buys $\lhd$-closedness.

The fibrational approach (3), beyond adjudicating the other two, yields a second thread: over **Set** it is the bifibration $\mathbf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$ of proof-relevant predicates on positions, whose fibred quantifiers are the All and Exists liftings and whose hyperdoctrine is the fibrewise opposite of the standard one — container-$\exists$ is **Set**-$\forall$, and each fibre is a co-topos. We developed this only at the position level; its shape-level quantifiers, the joint Beck–Chevalley and Frobenius laws, and the intrinsic class of Beck–Chevalley squares (Remark 6.7) remain open, and are the natural next targets on that thread.

We close by naming the single sharpest problem the survey surfaces, to seed the next proof.

Question 9.1 (Does the branching hierarchy lift index-wise?)**.** De Pascalis, Uustalu and Veltrì [6] classify the monoid structures on *indexed* containers over **Set**, dropping the cartesianness of [8] and separating a cartesian from a general regime. Over the single index their composition monad is exactly the substitution monoid $M \lhd (-)$ studied here. The external approach carries a finer, four-level branching hierarchy — λ-invertible $\subsetneq$ non-branching $\subsetneq$ cartesian $\subsetneq$ Π-Mendler — for the liftings of $M \lhd (-)$. *Does this four-level chain lift index-wise to the indexed containers* IC_I *of* [6]*, refining their cartesian/general dichotomy into a four-step tower over each index?* A positive answer would join the external approach (1) to the indexed approach (2) at the level of composition structure, exactly where Section 5 left them disjoint, and would be the first structural bridge between the two over a base richer than **Set**.

A second, more speculative target is whether Walker's bicategory of subcartesian polynomials ([15], Thm. 5.2.8), built from Street-style spans, is the same object as the external $\otimes/\lhd$ constructions of Section 4, built from family-style external coproducts, or genuinely disjoint — the direct overlap check for the container-over-**Vec** programme. We record it as secondary; Question 9.1 is the one to prove first.

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